

STATISTICAL VALIDATION OF 19-LETTER VERSES IN THE QURAN: A QUANTITATIVE STUDY OF THE NUMERICAL STRUCTURE

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ABSTRACT

Background: Mathematics represents the fundamental universal truth and language of the cosmos. Numerical patterns inherent in naturally occurring perfect formations are similarly evident in the Quran as a result of its divine authorship. A sacred text believed to be divine is expected to exhibit inherent proofs that can be verified to validate its authenticity, with mathematical proof being the most unambiguous. The unique textual integrity of the Quran, meticulously preserved through oral and written traditions in a specific order, renders it exceptionally suitable for numerical analysis. However, previous methods have been proposed subjectively and lack objective statistical testing.

Methods: This study is the first to use robust statistical tests to examine numerical patterns in the Quran by converting each letter to a Microsoft Excel cell. We analyzed the frequency of 19-letter verses in the Quran using a dataset of 6,236 verses with letter counts ranging from 3 to 551. Among other numerical patterns, we identified an outlier of 168 verses with 19 letters in our scatter plot analysis, prompting us to question whether this could occur randomly. Both chi-square goodness-of-fit tests and Poisson tests were applied. STATA 18.0 was employed for statistical analysis.

Results: An expected frequency of 70 from the empirical distribution was compared with the observed frequency of 168. The chi-square test yielded a statistic of 138.76 ($df = 1$, $p < 0.001$), and the Poisson test produced a Z-score of 11.71 ($p \approx 0$, using normal approximation). Both tests yielded highly significant results. The strong rejection of the null hypothesis by both tests demonstrates the statistically significant frequency of 19-letter verses (168 instances), which is highly improbable by chance. Quranic verses show a non-random distribution of letter counts, suggesting underlying structural or symbolic meaning and prompting further research into numerical patterns within sacred texts. The number 19, as mentioned in Quran 74:30-31, serves as a divine test to distinguish believers from disbelievers, affirming the Quran's divine origin and refuting the hypothesis that it consists merely of human words (74:25) while strengthening the faith of the People of the Book and true believers. The hypothesis that Prophet Muhammad or his companions intentionally designed the Quran's 19-letter verse frequency over 23 years is implausible due to their lack of mathematical expertise, focus on oral preservation, absence of historical significance for the number 19, and inconsistent numerical patterns across Quranic metrics.

Conclusion: The probability of the Quran being created by humans is approximately $1:1.73 \times 10^{31}$, an event comparable to locating a single grain of sugar within a mountain-sized pile, confirming its divine authorship.

Keywords: Religious texts, Holy Quran, Numerical structure, Nineteen

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1. Introduction

Islamic theology, linguistics, and mathematics are among the many fields that have extensively studied the Quran, Islam's central religious text. Mathematics, because of its ability to describe patterns, relationships, and structures in both physical and abstract realms without human perception or cultural influence, is a universal truth and the cosmos' foundational language.¹ Mathematics provides axioms that apply across the entire universe with no exceptions, unlike other sciences, which may depend on time, place, or subjective interpretation. Mathematics, with its abstract concepts and universal principles, serves as a powerful tool to unlock the mysteries of nature, from the orbits of planets to the behavior of subatomic particles.² Divine and natural manifestations of mathematical principles, such as π and the golden ratio, reveal mathematical patterns that are consistent and still being explored. Hence, numerical tests offer the best way to confirm hypotheses, as they are objective and not subject to conjecture, unlike other arguments. On the other hand, falsehood is accompanied by randomness and chaos. The Holy Quran's linguistic structure demonstrates profound thematic connectivity, weaving intricate layers of meaning through repetition, lexical cohesion, and rhetorical devices.³ The numerical patterns in the text of the Quran are intriguing, possibly holding symbolic or structural meaning, as some scholars and researchers suggest. Previous works on Letter 7 have also been conducted. The number 19, specifically referenced in Surah Al-Muddathir (74:30) as the number of angels guarding Hell ("Over it are nineteen"), has attracted significant scholarly attention. Scholarly examination of this verse has resulted in detailed research of the incidence of the number 19 throughout the Quran's textual organization, involving its appearance in the counts of letters, words, and verses.

Although there may be multiple numerical patterns, this paper focuses on a specific numerical observation: the frequency of verses in the Quran that contain exactly 19 letters. The objective of this study is to determine whether this frequency of 168 verses with 19 letters could occur by random chance or if it represents a statistically significant pattern. To address this, we employed two robust statistical tests: the chi-square goodness-of-fit test and the Poisson test. These tests compare the observed frequency against an expected frequency derived from the empirical distribution of letter counts. The results provide insight into the likelihood of this pattern occurring randomly and contribute to the broader discussion of numerical significance in the Quran.

¹ Eugene P. Wigner, "The Unreasonable Effectiveness of Mathematics in the Natural Sciences. Richard Courant Lecture in Mathematical Sciences Delivered at New York University, May 11, 1959," *Communications on Pure and Applied Mathematics* 13, no. 1 (1960), <https://doi.org/https://doi.org/10.1002/cpa.3160130102>.

² Mario Livio, *Is God a Mathematician?* (Simon & Schuster, 2009), <https://www.simonandschuster.com/books/Is-God-a-Mathematician/Mario-Livio/9780743294065>.

³ H. Khajehei and A. Shakarami, "An in-Depth Linguistic Analysis of the Thematic Connectivity in the Holy Quran," *Academic Research International* 3, no. 2 (2012), [http://savap.org.pk/journals/ARInt./Vol.3\(2\)/2012\(3.2-07\).pdf](http://savap.org.pk/journals/ARInt./Vol.3(2)/2012(3.2-07).pdf).

Black swan analogy: Statistics use the black swan as an illustration of inductive reasoning's fallibility. Seeing only white swans incorrectly suggested that all swans were white—a conclusion proven false by finding black swans in Australia. A key point is the importance of falsifiability; a universal hypothesis can be rejected with just one counterexample, highlighting the dangers of overgeneralizing from limited data.⁴

2. Methodology

2.1 Data description

The text of the Quran was imported from Tanzil, without pause marks and sajdah signs in plain text form.⁵ Tanzil is a Quranic project launched in 2007 to produce a highly verified Unicode Quran text for use in Quranic websites and applications. The Tanzil Quran text (Medina Mushaf), currently the most authentic copy of the Holy Quran (narration of Hafs), is carefully derived from authoritative sources. The text was imported from the text file via Python to Excel 365.

The Excel sheet preparation involved a systematic methodology to create a structured dataset of Quranic verses (ayat) for linguistic and statistical analysis. The process began by defining the objective to capture the Surah number, verse number, letter count, word count, Arabic text, and character-by-character breakdown using the Uthmani script as the primary source.

Data were extracted by segmenting the Quranic text into verses, computing letter and word counts (excluding diacritics), and decomposing each verse into individual characters. These attributes were organized into an Excel workbook, with each row representing a verse and columns storing the respective data, including sequential character columns. Automated scripts (e.g., Python with pandas) facilitated text processing, while validation ensured textual accuracy and correct counts through cross-referencing with authoritative sources and manual checks.

The methodology prioritized reproducibility, cultural sensitivity, and data integrity, producing a dataset suitable for scholarly analysis with potential for future extensions such as visualizations or additional attributes. Each letter was assigned a value according to the Abjad numerical system, and calculations were performed for each verse and chapter.

The dataset comprises 6,236 verses from the Quran, with each verse imported into Microsoft Excel 365. The columns adjacent to each verse display separate letters and words. Character counts were determined excluding spaces and diacritical marks, with word counts recorded in the adjoining column. EndNote 25.0 was used for reference management. STATA 18.0 (StataCorp, College Station, TX) was used for statistical calculations.

2.2 Statistical tests

⁴ Nassim Nicholas, "The Black Swan: The Impact of the Highly Improbable," *Journal of the Management Training Institut* 36, no. 3 (2008).

⁵ "Tanzil - Quran Navigator | Download," <https://tanzil.net/download/>.

The chi-square goodness-of-fit test and the Poisson test were used to assess the statistical significance of the occurrence of 168 nineteen-letter verses. These tests are suitable for examining frequency data and assessing whether an observed frequency significantly differs from an expected frequency under a null hypothesis of random occurrence.

2.2.1 Chi-Square Goodness-of-Fit Test

The chi-square goodness-of-fit test evaluates whether the observed frequency distribution of a categorical variable matches an expected distribution. Here, we categorized verses into two groups: those with 19 letters and those with any other letter count. The hypotheses are as follows:

Null hypothesis (H_0): The frequency of verses with 19 letters follows the expected distribution (i.e., it occurs by chance).

Alternative hypothesis (H_1): The frequency of verses with 19 letters does not follow the expected distribution (i.e., it is not due to chance).

2.2.2 Poisson Test

The Poisson test evaluates whether the observed frequency of an event follows a Poisson distribution, which is suitable for modeling discrete, non-negative counts such as the frequency of verses with a specific letter count. Given the right-skewed nature of the distribution, a Poisson model is a reasonable approximation. The hypotheses are as follows:

Null hypothesis (H_0): The frequency of verses with 19 letters follows a Poisson distribution with mean λ .

Alternative hypothesis (H_1): The frequency of verses with 19 letters does not follow a Poisson distribution with mean λ (specifically, it is higher than expected).

3. Results

The Quran is an extensive text. General statistics are presented in Table 1. The total number of letters in the dataset ranges from 3 to 551, with the longest verse (551 letters) found in Surah Al-Baqarah (2:182). The distribution of letter counts is right-skewed, with most verses having fewer letters, making the frequency of 168 verses with 19 letters unusually high.

A scatter plot of the number of letters in each verse, based on a dataset of 6,236 verses, revealed that 168 verses contain exactly 19 letters, representing a notable outlier. There are three surahs with 19 verses. The letter counts with total frequency distribution were visualized in a scatter plot. The plot shows a right-skewed distribution where most verses have fewer letters (clustered between 3 and 100), and the frequency decreases as letter count increases. A notable outlier is observed at 168 verses with exactly 19 letters, which is significantly higher than the frequency of verses with similar letter counts (e.g., 18 or 20 letters).

A scatter plot depicting word count versus frequency was also generated, indicating a frequency of 144 for 19 words (Figure 2). The Abjad calculation was performed by assigning numbers to letters as given in the supplement.

3.1 Chi-square goodness-of-fit test

The observed frequency of verses with 19 letters is 168, whereas the expected frequency, on the basis of the empirical distribution, is 70. The remaining verses (those with letter counts other than 19) total 6068, with an expected frequency of 6166. With 1 degree of freedom (two categories minus 1), the p value for $\chi^2 = 138.76$ is less than 0.001, leading to the rejection of the null hypothesis. This finding indicates that the frequency of 168 verses with 19 letters is statistically significant and unlikely to occur by chance.

For verses with the letter 19:

$$\frac{(168-70)^2}{70} = \frac{(98)^2}{70} = \frac{9604}{70} \approx 137.20$$

For verses with letter counts other than 19:

$$\frac{(6068-6166)^2}{6166} = \frac{(-98)^2}{6166} = \frac{9604}{6166} \approx 1.38$$

$$\chi^2 = 137.20 + 1.38 = 138.76$$

With 1 degree of freedom (two categories minus 1), the p value for $\chi^2 = 138.76$ is less than 0.001, leading to the rejection of the null hypothesis. The probability is approximately 1 in 1.73×10^{31} , an event comparable to locating a single grain of sugar within a mountain-sized pile.

Other special occurrences of number 19 in the text

Surah Fatiha contains three verses with 19 letters (1:1, 1:4, 1:5). These verses are of paramount importance as they form the basic foundation of monotheism. Verse 1:1, which is the initial part of every chapter, contains 19 letters. Verse 1:2 has 18 letters, but a similar final verse of Surah 37:182 has 19 letters. The most repeated verse (55:13) has 19 letters and is repeated 31 times. The total number of verses (114) and the Abjad of the Quran are perfect multiples of 19. Verses related to victory contain 19 letters (48:1, 3).

Surah Nasr (110) contains 19 words, while Surah Balad (90) is particularly special as it contains 361 letters, a perfect square of 19. A pivotal chapter in the Quran emphasizing monotheism, considered in hadith to be equivalent to one-third of the Quran, contains 19 words and 66 letters, an Abjad value corresponding to "Allah" (Table 3).

3.2 Poisson test

Given a Poisson distribution with a mean of 70 for 19-letter verses, the observed count was 168. Since λ is a large value (70), a normal approximation is used to determine the standard deviations. With a Z score of 11.71, the p value is virtually zero, meaning that the observed frequency is highly improbable given a Poisson distribution with a mean of 70. The observed frequency is statistically significant, as the null hypothesis is rejected.

$$Z = \frac{(168 - 70)}{\sqrt{70}} = \frac{98}{8.37} \approx 11.71$$

With a Z score of 11.71, the p value is essentially zero, indicating that the observed frequency is highly improbable given a Poisson distribution with a mean of 70. The

observed frequency is statistically significant, as evidenced by the rejection of the null hypothesis.

4. Discussion

Mathematical patterns and symmetries, indicative of universal truths, are inherent characteristics of any entity or system exhibiting structural or functional perfection. Geometric shapes such as the perfect triangle or square embody precise mathematical relationships (e.g., the Pythagorean theorem or equal side lengths), which are constitutive of their idealized forms.⁶ Any two lines can be proven parallel via simple mathematical axioms. Similarly, a scripture considered divinely perfect, such as the Quran, may manifest numerical patterns—such as consistent word frequencies, structural symmetries, or significant numerical constants (e.g., the number 19)—as a hallmark of its intrinsic order and harmony. The golden ratio found in nature is a perfect example of natural mathematics, as is Euler's constant. As an applied mathematics field, statistics offers a way to definitively verify or refute hypotheses. The idea that a true scripture must exhibit statistical and mathematical uniqueness to prove its truth is a modern approach to validating sacred texts, often tied to claims of divine origin through patterns or codes.

4.1 The Role of the Omnipresent Observer

The universe operates under clear scientific rules, yet its intricate order suggests purposeful design. Newton's laws demonstrate that no action occurs without external force, and the second law of thermodynamics reveals that disorder in the universe continues to increase. However, we observe the opposite around us—energy is harnessed efficiently to perform remarkable tasks, from the smallest bacteria to complex humans, all powered by similar mitochondria.

Humans, despite their intelligence, are inherently limited. For example, they can only hear sounds between 20 Hz and 20,000 Hz and see a small fraction of the electromagnetic spectrum (visible light). These limitations mean that human observations are relative, shaped by imperfect senses, and cannot fully grasp universal truths without guidance.

4.2 Scripture and Verifiability

The existence of a Creator logically implies the necessity of Scripture. Flawless guidance is exclusively attributable to an omnipresent entity—a singular Creator offering divine wisdom through a single sacred text for humanity's refinement. The property of this ideal divine scripture is that it should be verifiable as divine to differentiate such claims from false or adulterated texts that exploit humanity. The Quran contains several verification mechanisms; however, we focus on a robust statistical approach that is universally accepted.

⁶ J. R. B. Brown, *Philosophy of Mathematics: A Contemporary Introduction to the World of Proofs and Pictures* (Routledge, 2010).

4.3 Quranic Numerical Analysis

The Quran is uniquely suited for numerical analysis due to its singular, unaltered version—the only book with a single written text and no variant editions. It has been meticulously preserved for centuries through both oral transmission and written records, with no alterations ever claimed. The Quran's memorization on such a massive scale is unparalleled, with millions of memorizers preserving its text across generations.⁷ This practice highlights Islam's deep emphasis on oral tradition and scripture preservation, fulfilling God's claim that He will preserve it.

Quran memorization dates to Prophet Muhammad's time (7th century CE) to preserve the text orally before widespread writing became available. It is often performed in original Arabic, even by non-Arabic speakers. Its rigorous organization and specific order further establish it as an ideal document for such research. Other scriptures lack such strong preservation, have multiple versions, and are not in their original form or language.

This paper's uniqueness lies in its organization of the Quran's letters as numbers, placed in separate Excel cells so that mathematical operations can be performed. Professional software (STATA 18.0), exhaustive verse-by-verse letter and word counts, comprehensive Abjad calculations (24,401,973, divisible by 19), and high-quality scatter plots (Figures 1 and 2) were employed. No previous study has undertaken such detailed, data-driven analysis of the Quran's numerical structure, nor has any applied such robust statistical tests to validate non-randomness. The paper's meticulous and clear presentation makes it a groundbreaking and scientifically rigorous contribution to Quranic studies, strongly supporting the hypothesis of intentional, non-random design.

The results of both the chi-square goodness-of-fit test and the Poisson test strongly suggest that the frequency of 168 verses with exactly 19 letters in the Quran is not random. The chi-square test statistic of 138.76 ($p < 0.001$) and the Poisson test Z-score of 11.71 ($p \approx 0$) provide compelling evidence that this frequency deviates significantly from what would be expected under random distribution.

Some scholars argue that 19 serves as a mathematical code embedded in the Quran, potentially acting as a mechanism for verifying the text's authenticity. Previous work has identified other patterns involving the number 19, such as the frequency of certain letters and the structure of specific surahs. The highly significant results from two robust statistical tests in this study offer objective support for further investigation into the number 19's non-random presence in the Quran.

4.4 Statistical Analysis and Previous Research

The numerical miracle, as explored by scholars such as Abdul Razzaq Nawfal, is presented as evidence of the Quran's divine origin, emphasizing word repetitions, structural patterns, and numerical relationships that align with natural or religious phenomena. He counted specific words (e.g., "day," "month," "prayers") and linked their frequencies to natural or religious phenomena. Highlighting the precision of these patterns as a sign of divine authorship, the authors argue that such consistency may not be coincidental or crafted by a

human, especially in an oral tradition.⁷ Previous works had involved numbers 7 and 19 but have relied on descriptive counts (e.g., word repetitions, letter frequencies) without statistical tests to confirm nonrandomness.⁸ A new study shows that the number 10 is significant in Surah al-Kawthar, appearing in word counts, letter frequencies, and symbolic links to Islamic practices, hinting at a deliberate, mathematically exact structure.⁹ Halim Sayoud's 2015 study concluded that the Quran contains a complex 7-based numerical structure identified through linguistic and statistical analysis, suggesting divine authorship due to its sophistication and improbability of human origin.¹⁰ A paper highlights examples such as the gematical value of "al-hadid" (iron) equaling 57, matching the 57th chapter's position, and "hadid" (iron) equaling 26, the atomic number of iron, and the word "night" appearing 92 times.¹¹

Another analysis incorporates the linguistic, contextual, and theological perspectives of the numerical and "Day" concepts within Surahs 32 and 70 to demonstrate their role in bolstering believers' confidence in divine justice, resurrection, and accountability. This work emphasizes that the numbers are not merely quantitative but also carry qualitative, symbolic weights, enhancing the divine message.¹² Recently, advancements have been made to explore the potential of Qur'anic natural language processing (NLP) to bridge traditional scholarship with modern technology, offering tools that respect the text's sacredness while enhancing its study, accessibility, and application for diverse global communities.¹³

Therefore, earlier works often framed numerical patterns within a theological narrative, assuming divine origin without empirical testing. Our use of the chi-square and Poisson tests introduces a scientific framework, making it the first to provide statistically significant evidence supporting the nonrandomness of a specific Quranic pattern.

4.5 The Significance of Number 19

One interpretation of the high frequency of verses with 19 letters is that they reflect an underlying structural feature of the Quran's composition. The Quran was revealed in Arabic, a language with a rich morphological and phonetic system, and its verses vary

⁷ A. R. Nofal, *The Numerical Miracle of the Holy Quran*, vol 5 (Arab Book House, 1987).

⁸ Imran Ali, "The Mathematical Miracles of the Holy Qur'an," *Ulūm al-Sunnah* 1, no. 01 (2023).

⁹ MZ Alghar and A. Rizqiyah, "Integration of Islam and Mathematics: A Study of the Number 10 in Surah Al-Kautsar," *Muhajirin International Journal* (2025), <https://ejournal.taqaddum.co.id/index.php/mij/article/view/1234>.

¹⁰ H. Sayoud, "New Numerical Hidden Structure in the Holy Quran Based on Number 7," *HDSKD International Journal* (2015), http://sayoud.net/Number_7_Quran.pdf.

¹¹ Magazine The Fountain, "Numerical Codes and Gematrical Mysteries in the Qur'an," *The Fountain Magazine* (2015), <https://fountainmagazine.com/2015/issue-101-may-june-2015/numerical-codes-and-gemetrical-mysteries-in-the-quran>.

¹² K. H. Hussein, "The Meaning of the Day and Numbers, Which Are Included in Surat Al-Sajdah and and Surat Al-Ma'araj," *Iraqi University Journal* 45, no. 1 (2019).

¹³ Muhammad Huzaifa Bashir et al., "Arabic Natural Language Processing for Qur'anic Research: A Systematic Review," *Artificial Intelligence Review* 56, no. 7 (2023), <https://doi.org/10.1007/s10462-022-10313-2>.

widely in length and content. Word choice, grammar, and rhetorical style all affect verse length, and these elements may have been deliberately designed to create patterns.

Another interpretation is symbolic. The presence of 19 in Surah Al-Muddathir (74:30, 31) has led some to believe it holds great religious significance in Islam, representing divine order, as the subsequent verse calls it a test of faith.

The number 19 is special in both mathematics and nature. It is prime and a centered hexagonal number, fitting into geometric patterns such as hexagons, which appear in structures like crystals. In nature, 19 is close to the Fibonacci number 21, so it might appear in plant patterns such as petal numbers or seed spirals in flowers, which often follow Fibonacci-like sequences.¹⁵ Its prime nature could also relate to natural cycles, such as those of certain insects, helping them avoid predators. Additionally, 19 often appears in patterns such as the 19-year Metonic cycle, which aligns lunar and solar calendars. These remain speculative; the reason for Allah's choice of 19 is revealed in Surah 74 of the Quran:

"It is overseen by nineteen [keepers]. We have appointed only [stern] angels as wardens of the Fire. In addition, we have made their number only as a test for the disbelievers, so that the People of the Book will be certain, and the believers will increase in faith, and neither the People of the Book nor the believers will have any doubts, so that those [hypocrites] with sickness in their hearts and the disbelievers will argue, 'What does Allah mean by such a number?' In this way, Allah leaves whoever He wills to stray and guides whoever He wills. In addition, no one knows the forces of your Lord except Him. In addition, this [description of Hell] is only a reminder to humanity."

Verse 74:25 of Surah Al-Muddathir refutes non-believers' assertion that this is simply human speech; creating such a definite pattern would not have been possible for humans. Another verse states: "Had it been from [any] other than Allah, they would have found within it much contradiction" (4:82). Additionally, Allah guarantees that He will preserve it (15:9, 75:17-20) and challenges disbelievers to produce a single Surah like the Quran, telling them they will never be able to do so (2:23-24, 11:13, 17:88).

The conjecture that the Quran's frequency of 19-letter verses was intentionally designed by Prophet Muhammad's companions over 23 years is implausible. The companions, who focused on memorizing and preserving the Quran's message within an oral tradition, were not mathematicians capable of such rigorous calculations over such an extended period. No historical evidence suggests that 19 held special significance to them, unlike numbers such as 7 or 12, which carry cultural weight in Abrahamic traditions, raising the question of why they would prioritize 19. Furthermore, the inconsistent behavior of 19 across other Quranic metrics suggests that this pattern is unlikely to stem from human design, as intentional structuring would likely show broader numerical consistency.

4.6 Addressing Criticisms

From the time of the Quran's revelation to the present, some individuals have not accepted it as God's word. They alleged that he was being taught by humans (16:103) or that it was a combination of fabricated dreams. However, other evidence demonstrates that the Quran

could not have been crafted by an unlettered person of that time. The Prophet never traveled to the sea but claimed through the Quran that there were seas that did not mix (55:19-20). The Quran contains verses that align with modern scientific discoveries, such as embryology, the universe's expansion, and geological phenomena. These are often cited as evidence of its divine origin, as they appear consistent with facts established long after the 7th century. The Quran accurately details embryonic development (23:12-14), asserts the universe's expansion (51:47), posits that life began in water (21:30), describes everything as orbiting (36:40), and identifies mountains as stabilizers (78:6-7), among other claims. The Quran's mathematical and statistical uniqueness, as demonstrated by the significant frequency of 19-letter verses, underscores its extraordinary structural precision. The chi-square goodness-of-fit test ($\chi^2 = 138.76$, $p < 0.001$) and Poisson test ($Z = 11.71$, $p \approx 0$) reveal a non-random distribution of 168 verses with exactly 19 letters, a pattern highly improbable by chance. The probability is approximately 1 in 1.73×10^{31} , an event comparable to locating a single grain of sugar within a mountain-sized pile. This numerical consistency, along with other occurrences of the number 19 in surah counts, word frequencies, and Abjad values, suggests a deep mathematical structure resembling fundamental constants. Such patterns suggest intrinsic design that transcends human authorship, aligning with the Quran's claim of divine origin. These findings invite further exploration into the interplay of mathematics and sacred texts, reinforcing the notion that the Quran's composition embodies a unique, statistically significant framework warranting continued scholarly investigation.

4.7 Scientific Harmony

We anticipate that critics will try to undermine the strong statistical evidence by twisting specific passages, presenting them as scientific assertions while ignoring their real meaning. However, these verses are intended to provide overall direction regarding morality, spirituality, and ethics. For example, critics allege an error in inheritance shares sometimes exceeding unity (e.g., 4:11-12), but this is resolved through proportional adjustment ('awl), a principle derived from Quranic analogies on justice and debts, ensuring equitable distribution without contradiction. Another objection is the sun setting in a "muddy spring" (18:86), yet this employs phenomenological language describing visual perception, not literal geocentrism, as clarified by early commentators. Idiomatic expressions are sometimes employed to simplify complex subjects; the Quran itself advises against misinterpreting verses, something the ignorant will attempt (3:8). Idiomatic expressions are so common in everyday language that even now word sunset and sunrise are used. Claims of inaccurate embryology (23:12-14) overlook "alaaqah" as a metaphor for the leech-like clinging embryo, with non-linear Arabic phrasing aligning with modern stages. Similarly, semen from "between backbone and ribs" (86:6-7) refers to the emission region near the prostate, not production, consistent with anatomical paths. Mountains as "pegs" (78:6-7) accurately depict their deep roots stabilizing the crust via isostasy, per geology. The Quran is crafted to remain accessible and understandable to people of all ages,

cultures, and intellectual capacities, offering enduring principles that transcend temporal scientific paradigms rather than functioning as a technical manual for experimentation or proof in disciplines like physics or biology. This timeless quality underscores its divine authorship, as it harmonizes with advancing human knowledge without contradiction. Notably, prominent scientists have acknowledged the Quran's remarkable consonance with modern discoveries, bolstering its credibility. For instance, French surgeon Maurice Bucaille, in his seminal work "The Bible, the Qur'an and Science" (1976), concluded after extensive analysis that the Quran contains no statements conflicting with established scientific facts, attributing this to its supernatural origin.¹⁴ Likewise, Professor Keith L. Moore, a leading embryologist and author of "The Developing Human: Clinically Oriented Embryology" (multiple editions, with Islamic additions in 1983), has expressed astonishment at the Quran's precise descriptions of embryonic stages, stating that such knowledge, unknown in the 7th century, must derive from divine revelation, though some interpretations have evolved.¹⁵ These endorsements, among others, invite further interdisciplinary exploration into the Quran's multifaceted profundity.

4.8 Logical Fallacies in Quranic Criticism

Some individuals, whether through clever rhetoric or sheer ignorance, attempt to dispute the Quran's divinity by invoking historical events such as wars and bloodshed in early Islam. They suggest that these human conflicts undermine the scripture's divine origin, rejecting truth in the name of false logic, as Iblis did when he said, "I am better, as I am created from fire and Adam is created from clay."

This approach relies on a logical fallacy that distracts from the Quran's textual and theological claims. They employ the ad hominem fallacy, which involves attacking the character, circumstances, or associated elements of a claim's source (in this case, the historical context of Islam) rather than addressing the argument's merits directly. This distorts reasoning and leads to flawed conclusions, much like dismissing a mathematical truth such as $2 + 2 = 4$ simply because the teacher explaining it once failed a test—an ad hominem attack that ignores the fact's inherent validity by shifting focus.

The fallacy shifts focus to irrelevant personal or situational flaws to discredit the idea, avoiding substantive evaluation of evidence such as the Quran's linguistic precision, prophecies, or monotheistic teachings.

An everyday example of the ad hominem fallacy occurs when someone dismisses a smoker's warning about health risks by arguing that many famous people smoked and that smoking was culturally accepted at that time. This diverts attention from the argument's validity regarding smoking's dangers and the supporting medical evidence.

Similarly, rejecting the Pythagorean theorem because its originator and followers held bizarre beliefs—such as avoiding beans because they contained souls of the dead, shunning

¹⁴ Maurice Bucaille, *The Bible, the Qur'an and Science* (Seghers (original French edition), 1976).

¹⁵ Keith L. Moore et al., *The Developing Human: Clinically Oriented Embryology* (W.B. Saunders Company (early editions); Elsevier (later editions), 1973).

public roads, always putting on the right shoe first, and worshipping numbers as divine entities—would be illogical. The theorem's validity stands independent of those superstitions.

In addition to the ad hominem fallacy, several other logical fallacies can be recognized and addressed when defending against flawed reasoning that distorts truth. The straw man fallacy misrepresents an opponent's position to make it easier to refute. The slippery slope fallacy claims that a small step will inevitably lead to extreme consequences without providing evidence. Appeal to authority relies on an authority figure's opinion instead of valid evidence. A false dilemma presents only two options when more exist, forcing a choice between extremes.

These fallacies are often employed by those who consider themselves intellectually superior when arguing against the Quran's truth. To avoid being misled by such flawed reasoning, one should critically analyze arguments by identifying logical errors, seeking verifiable evidence from multiple reliable sources, and applying sound logic while remaining open to counterarguments and self-reflection.

The Quran itself addresses one common fallacious argument—the appeal to tradition—by refuting the claim that we should simply follow our ancestors' faith. It argues that those ancestors were not necessarily more knowledgeable or correct in their beliefs.

4.9 Quranic Perspective on Persistent Disbelief

The Quran frequently highlights that some disbelievers reject faith regardless of extraordinary signs or miracles, such as raising the dead, moving mountains, or descending parchment from the sky. Key verses include Surah Al-Baqarah (2:6-7), which notes sealed hearts unmoved by warnings; Surah Al-An'am (6:7, 6:111), which states that even tangible scripture or speaking dead would be dismissed as magic; Surah Ar-Ra'd (13:31), which asserts that disbelief persists even if the Quran could perform miracles such as cleaving the earth; and Surah Al-Isra (17:90-95), which lists demands for miracles such as gushing springs or heavenly books yet rejecting human messengers.

According to the Quran, envy, arrogance, pride, and the pursuit of material wealth or status are cited as reasons for disbelief. Believers' inability to reflect honestly, or their clouded judgment, prevents them from seeing truth, even when obvious; likewise, blindly following tradition prevents them from recognizing divine guidance. Despite these clear teachings, many would still ignore the proofs that believers consider evidence of the Quran's divine nature.

"Surely, We have revealed the 'Dhikr' (Quran), and surely, We will preserve it" (15:9) "No falsehood will come to it, in the present or the future; a revelation from One who is Wise and Praiseworthy" (41:42)

Conflict of Interest

None

Ethics Statement

This study analyzes the Holy Quran's numerical structure with utmost respect for its sacred status in Islam. No alterations or reinterpretations of the text were made beyond objective counts from verified sources (e.g., Tanzil.net). The work is intended for scholarly exploration of patterns, adhering to cultural sensitivity, reproducibility, and integrity, without involving human subjects. It complies with ethical guidelines in religious studies, emphasizing non-maleficence and respect for beliefs, with conclusions presented as hypotheses for discussion, not proselytization.

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Table 1: General statistics

Metric	Value
Chapters (Surahs)	114 (19×6)
Verses	6236
Words	78245
Letters	332837
Repeated verses >2 times	2232
Most repeated verse	31
Number of letters in most repeated verse	19
Total abjad value	$24401973 = 1284314 \times 19$
Number of Surah with 19 verses	3

Table 2: Frequency distribution of letters and words

Metric	Letters	Words
Sample Size (n)	2568	2568
Range	3–551	1–129

Mode	19 (168 occurrences)	9 (240 occurrences)
Mean (SD)	51.61 (46.98)	11.86 (8.67)
Median	35	10
Top 5 Frequent Values	19 (106), 18 (82), 17 (79), 20 (74), 16 (67)	9 (240), 8 (227), 7 (234), 10 (212), 11 (219)
Most repeated verse	19	4
Values with Frequency ≥ 50	16–24 (639 total)	5–13 (1903 total)

Table 3: Other instances of number 19 in the Quran

Surah No.	Verses Divisible by 19	Words Divisible by 19	Letters Divisible by 19	Abjad Value Divisible by 19
9	–	–	585	–
10	–	97	–	–
21	–	62	–	–
30	–	–	–	14,667
41	–	42	–	–
47	2	–	–	–
57	–	–	–	10,325
68	–	–	–	4,990
71	–	–	–	4,006
76	–	13	–	–
82	1	–	–	–
87	1	4	–	–
90	–	–	19*	1,319
95	–	2	–	–
96	1	4	–	–
107	–	–	7	526
111	–	–	–	348
112	–	1* (19)	–	Abjad value equal to 66 (Abjad value of “Allah”

*Square of 19

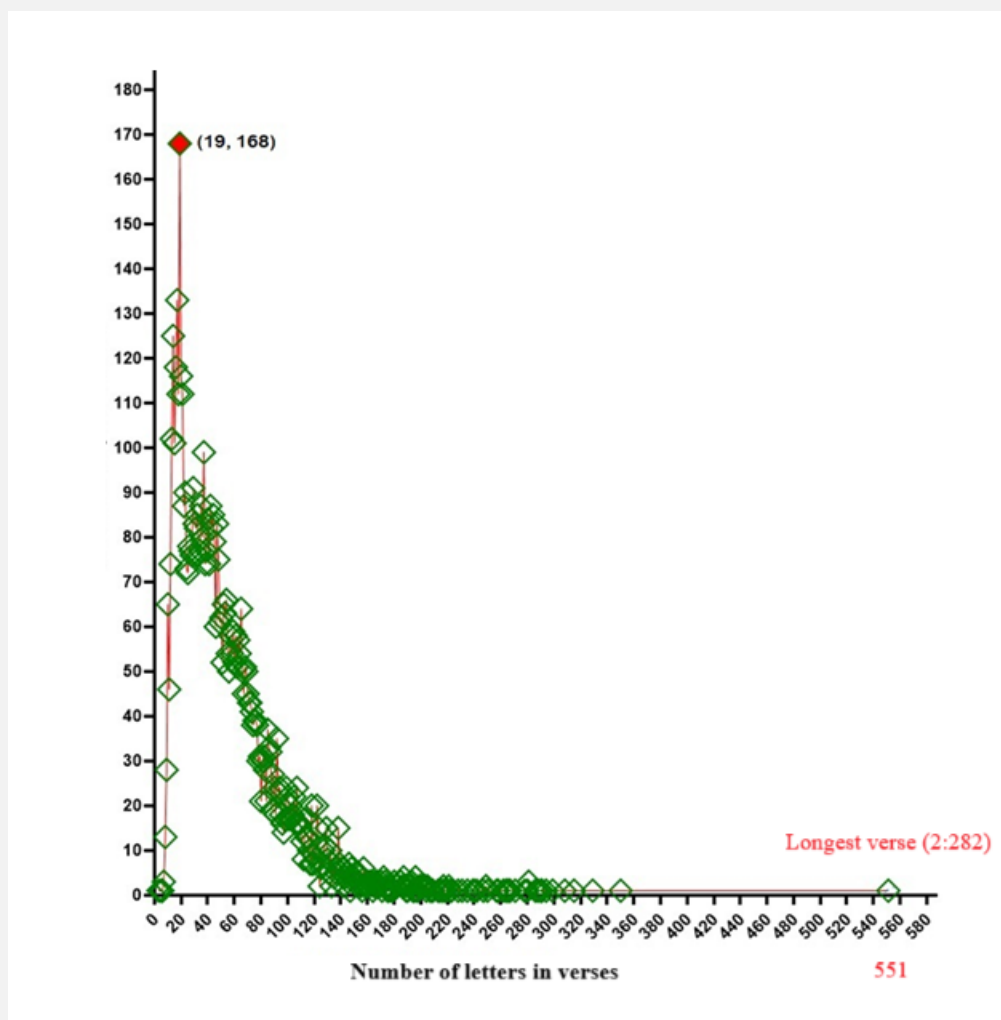


Figure 1: Total verses: 6236; Letters range: 3–551; X-axis: Number of letters in verse at (1–580); Y-axis: Frequency of verses (1–180); Green diamonds: Individual verses; Red dashed line: Trend of distribution; Brown diamond: Outlier at (19, 168); Longest verse: 2:182 with 551 letters at (551, 19).

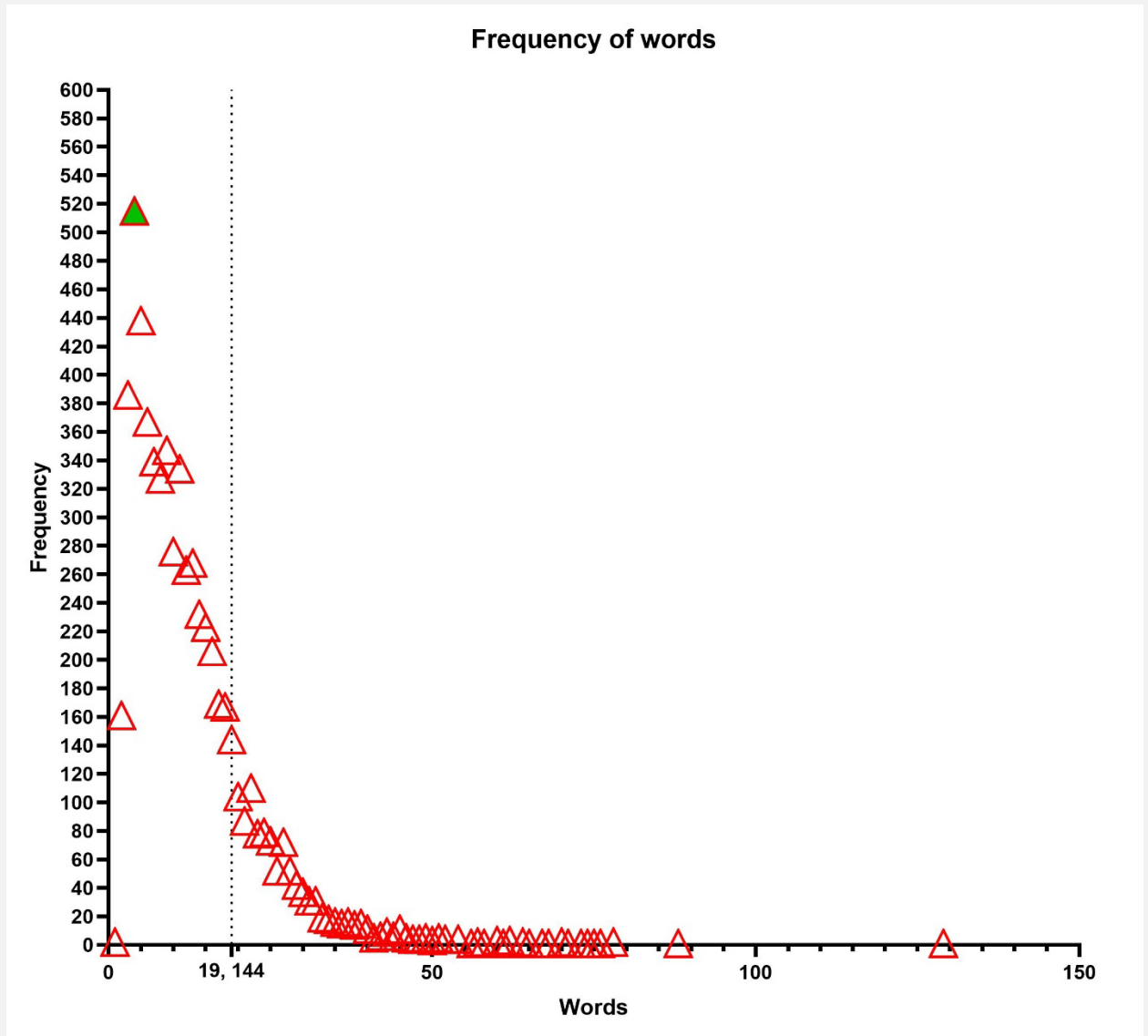


Figure 2: Number of words; Y-axis: frequency of words (1–600); red triangles: individual word frequencies; green triangles: (19, 144); distribution: right-skewed, most words below 50.

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Supplementary Table 1: Summary of chapters

Chapter	Number of verses	Total words	Total letters	Abjad
1	7	29	143	10147
2	286	6144	26268	1911634
3	200	3505	15004	1097171
4	176	3767	16351	1164030
5	120	2841	12225	859860
6	165	3060	12745	962327
7	206	3345	14454	1097079
8	75	1246	5406	388534
9	129	2505	11115	772555
10	109	1843	7608	553847
11	123	1950	7836	560302
12	111	1799	7326	511136
13	43	857	3564	250144
14	52	834	3558	271475
15	99	661	2901	187776
16	128	1848	7851	584999
17	111	1562	6662	494194
18	110	1587	6571	516191
19	98	975	3954	292234
20	135	1357	5418	402748
21	112	1178	5113	366020
22	78	1283	5333	382579
23	118	1056	4502	357237
24	64	1323	5773	435789
25	77	900	3897	302642
26	227	1324	5649	404365
27	93	1163	4809	350618
28	88	1442	5949	416771
29	69	982	4336	287651
30	60	821	3491	278673
31	34	554	2190	163366
32	30	376	1582	115189
33	73	1307	5807	392256
34	54	888	3613	263773
35	45	782	3257	253744
36	83	734	3087	227541

Chapter	Number of verses	Total words	Total letters	Abjad
37	182	869	3918	268122
38	88	739	3084	236948
39	75	1181	4888	378909
40	85	1230	5127	371366
41	54	798	3384	248893
42	53	864	3541	254780
43	89	840	3628	267036
44	59	350	1493	99506
45	37	492	2104	159785
46	35	649	2686	196354
47	38	546	2442	187380
48	29	564	2529	189859
49	18	357	1552	103776
50	45	377	1526	110882
51	60	364	1565	120538
52	49	316	1343	95059
53	62	364	1452	115759
54	55	346	1488	125541
55	78	356	1666	130539
56	96	383	1775	133744
57	29	579	2564	196175
58	22	479	2065	135342
59	24	451	1989	134908
60	13	356	1579	104050
61	14	230	985	64569
62	11	181	787	63582
63	11	185	820	56100
64	18	246	1110	80607
65	12	293	1222	93434
66	12	258	1124	88813
67	30	337	1366	104644
68	52	305	1308	94810
69	52	264	1152	97937
70	44	221	990	73756
71	28	231	984	76114
72	28	290	1128	68168
73	20	204	873	70754

Chapter	Number of verses	Total words	Total letters	Abjad
74	56	260	1054	92555
75	40	168	695	54594
76	31	247	1106	78663
77	50	185	860	72616
78	40	178	815	54279
79	46	183	804	68113
80	42	137	571	54137
81	29	108	454	42292
82	19	85	352	27463
83	36	173	769	57922
84	25	112	464	33759
85	22	113	488	33650
86	17	65	273	17778
87	19	76	315	27084
88	26	96	401	38226
89	30	143	603	49981
90	20	86	361	25061
91	15	58	272	20168
92	21	75	333	30414
93	11	44	184	14108
94	8	31	121	13202
95	8	38	181	11018
96	19	76	307	23325
97	5	34	134	9091
98	8	98	423	32446
99	8	40	177	18185
100	11	44	188	14157
101	11	40	179	13436
102	8	32	142	11163
103	3	18	92	5528
104	9	37	153	9904
105	5	27	116	6652
106	4	21	96	6850
107	7	29	133	9994
108	3	14	62	3550
109	6	31	118	4618
110	3	23	99	6910

Chapter	Number of verses	Total words	Total letters	Abjad
111	5	27	100	6612
112	4	19	66	1788
113	5	27	92	9463
114	6	24	99	6082
TOTAL	6236	78245	332837	24401973